Final Project Report

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CMSC 462

1. Data Description

Q1: The clean_weather_daily dataset has 367 rows and 13 columns.

Q2: Table of Column Variables and Types:

Column Variable	Type
STATION	Nominal
DATE	Datetime
PRCP	Continuous
SNOW	Continuous
SNWD	Continuous
TAVG	Continuous
TMAX	Continuous
TMIN	Continuous
AWND	Continuous
WDF2	Continuous
WDF5	Continuous
WSF2	Continuous
WSF5	Continuous

2. Data Validation & Cleaning

Q3: Table for Missing Values for Each Variable:

Column Variable	Missing (#)	Missing (%)
STATION	0	0.00%
DATE	0	0.00%
PRCP	0	0.00%
SNOW	1	0.27%
SNWD	19	5.18%
TAVG	95	25.89%
TMAX	0	0.00%
TMIN	0	0.00%
AWND	96	26.16%
WDF2	96	26.16%
WDF5	96	26.16%
WSF2	96	26.16%
WSF5	96	26.16%

SNOW - It is likely MCAR since there is only a single random missing data point, which could reasonably be the result of a clerical or recording error.

SNWD - It is likely MAR because values were missing even on days where snowfall was recorded. The missingness may be related to the reporting process or measurement issues on those specific days.

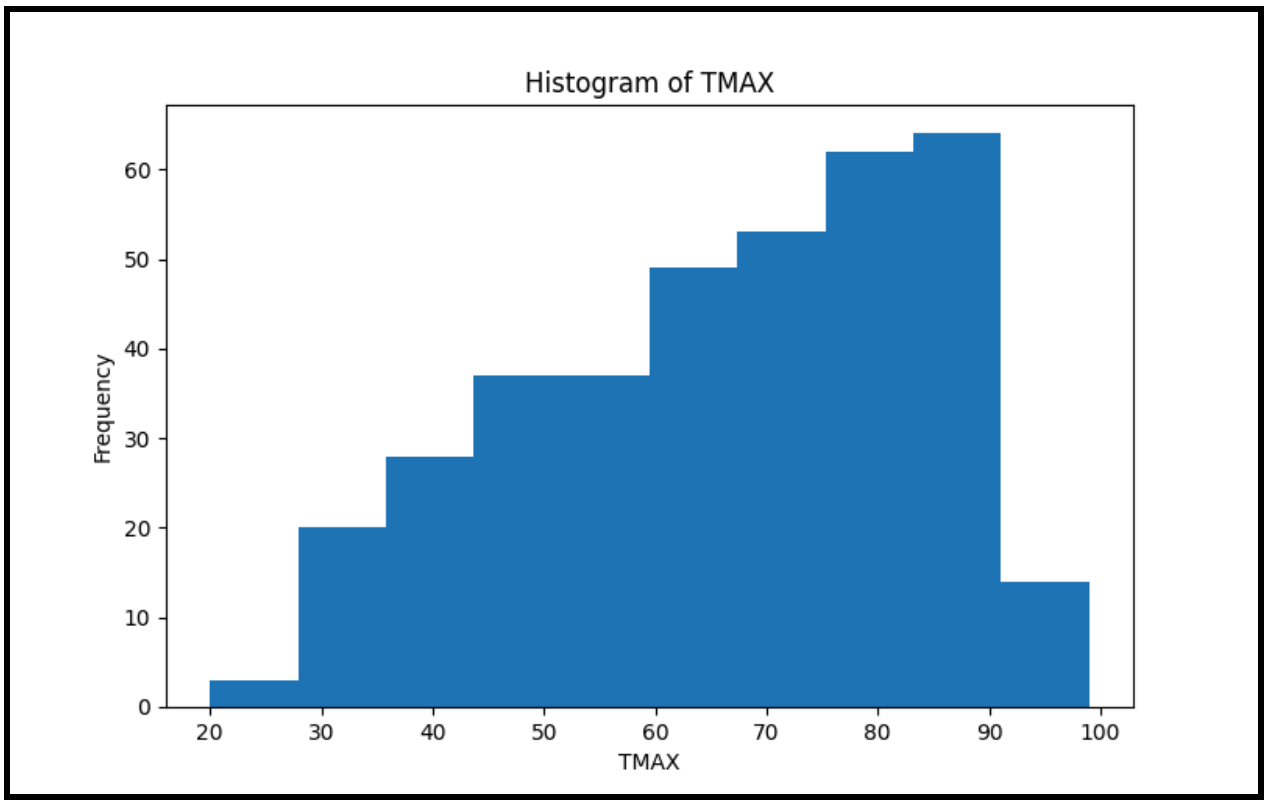
TAVG - It is likely MAR because the missingness occurs as one continuous block starting from August 28th, 2025, through the end of the year. Additionally, based on the corresponding TMIN and TMAX values, the temperatures during that period were not unusually extreme compared to earlier recorded values.

All Wind Variables - They are all likely MAR because the missingness occurs as one continuous block beginning on August 27th, 2025. This pattern suggests the missingness may be related to station maintenance, reporting interruptions, or equipment malfunction rather than the wind measurements themselves.

Q4: I did not observe any duplicate DATE values in the dataset. I checked for duplicate DATE values by using this line of code: `df1["DATE"].duplicated().any()`, from the pandas python library.

3. Exploratory Data Analysis

F1 - Histogram of TMAX

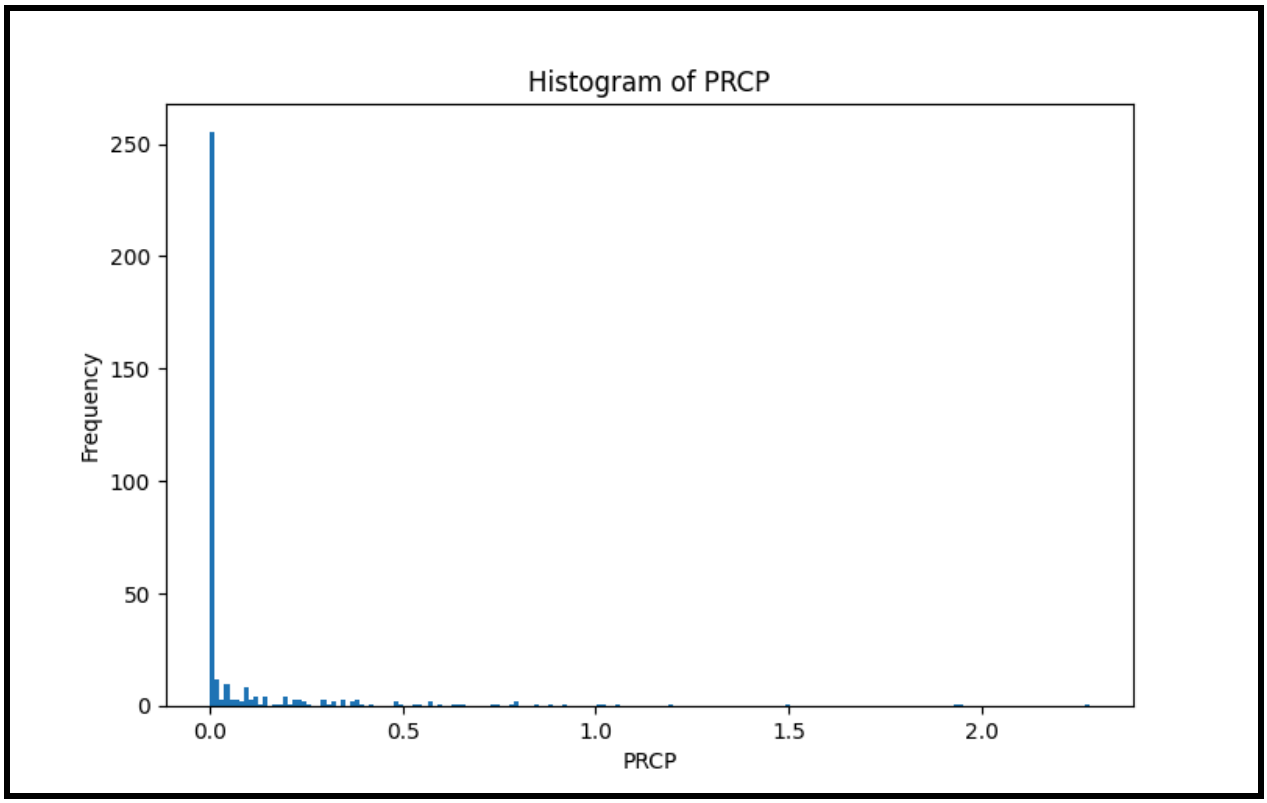


F1-Q5: This histogram for the variable of TMAX is using a number of bins determined by Freedman-Diaconis Rule. The shape of the histogram looks to be unimodal and moderately left-skewed.

F1-Q6: There does not seem to be any visible outliers in this histogram.

F1-Q7: Something that stands out within the histogram is that it shows a high concentration of temperatures in the higher end. Most of the observations are clustered around the 65 - 90 degrees Fahrenheit range.

F2 - Histogram of PRCP

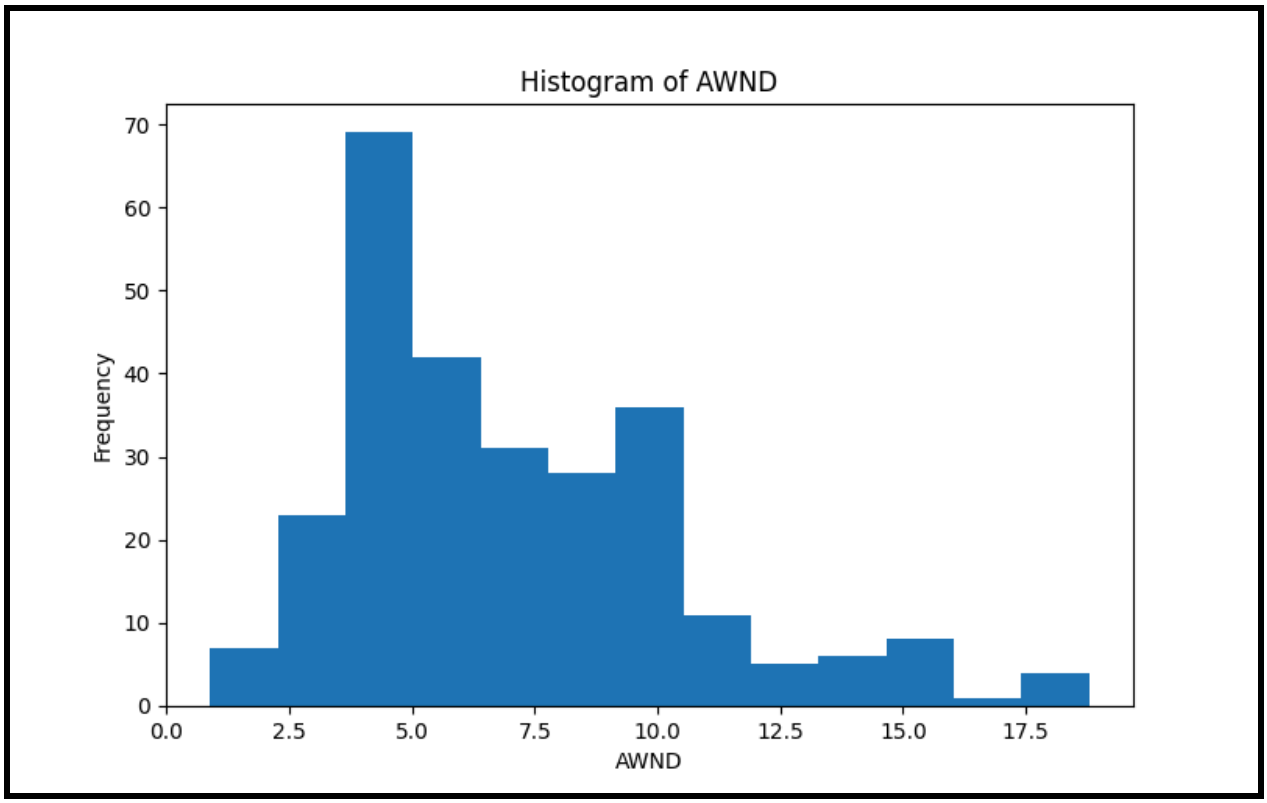


F2-Q5: This histogram for the variable of PRCP is using a number of bins determined by Freedman-Diaconis Rule. The shape of the histogram looks to have a heavy right-skew distribution.

F2-Q6: There appear to be several outliers at approximately 0.5 inches of precipitation and higher, representing days with unusually heavy rainfall compared to the majority of observations.

F2-Q7: One surprising observation is that more than 250 days recorded 0 inches of precipitation. This means that approximately 69% of the 367 days in the dataset experienced no precipitation at all.

F3 - Histogram of AWND



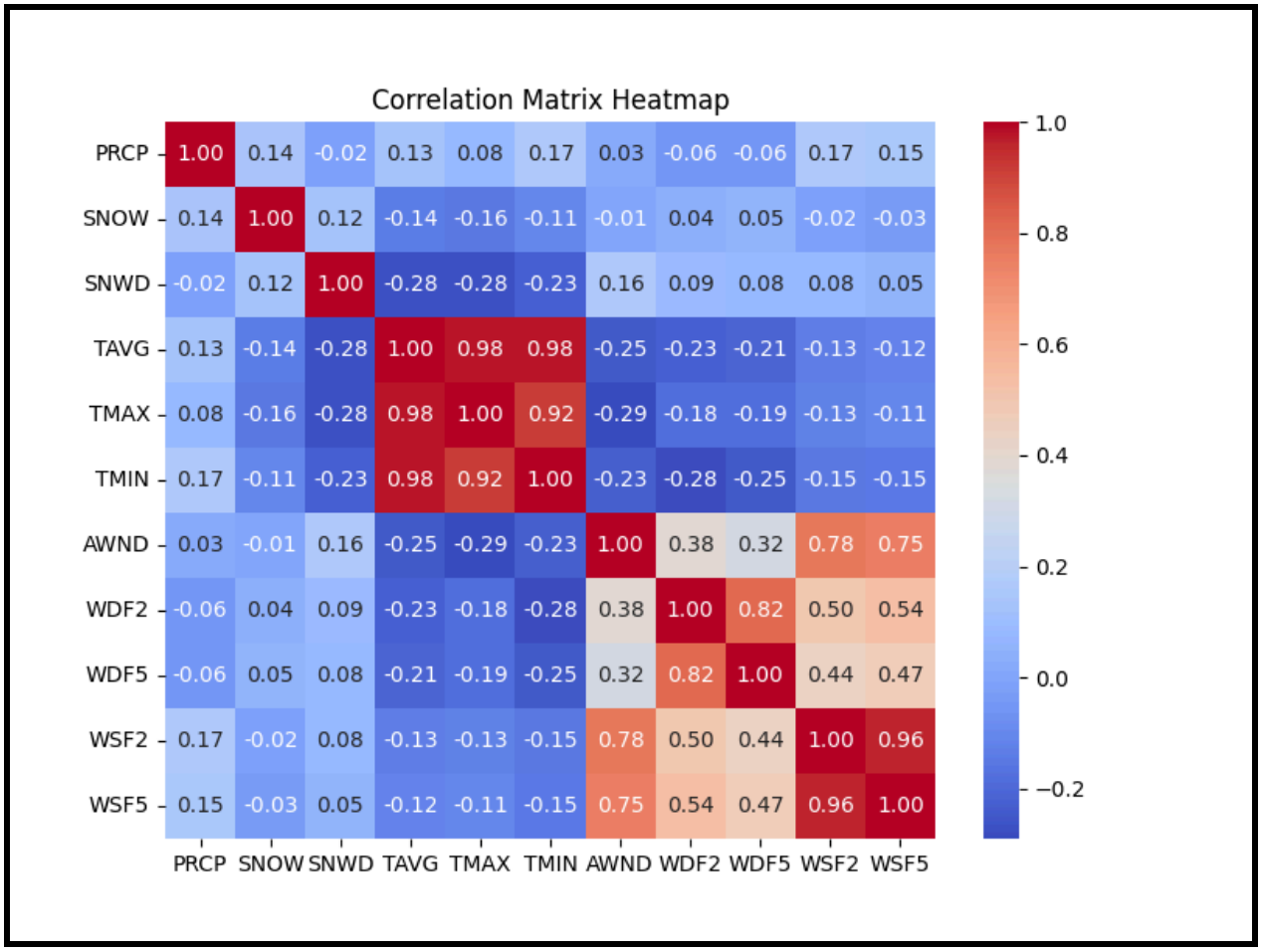
F3-Q5: This histogram for the variable of AWND is using a number of bins determined by Freedman-Diaconis Rule. The distribution appears to be moderately right-skewed. Most wind speed observations are concentrated between about 3 and 10, with fewer observations at higher wind speeds creating a longer tail on the right side of the histogram.

F3-Q6: There appear to be a few potential outliers at the higher end of the distribution, particularly around 15 to 19, representing unusually windy days compared to the majority of observations.

F3-Q7: One interesting observation is that the wind speed values are spread across a fairly wide range rather than being tightly clustered around a single value. Additionally, extremely high

wind speeds are relatively rare compared to the more moderate wind speeds observed throughout the dataset.

F4 - Correlation Matrix Heatmap



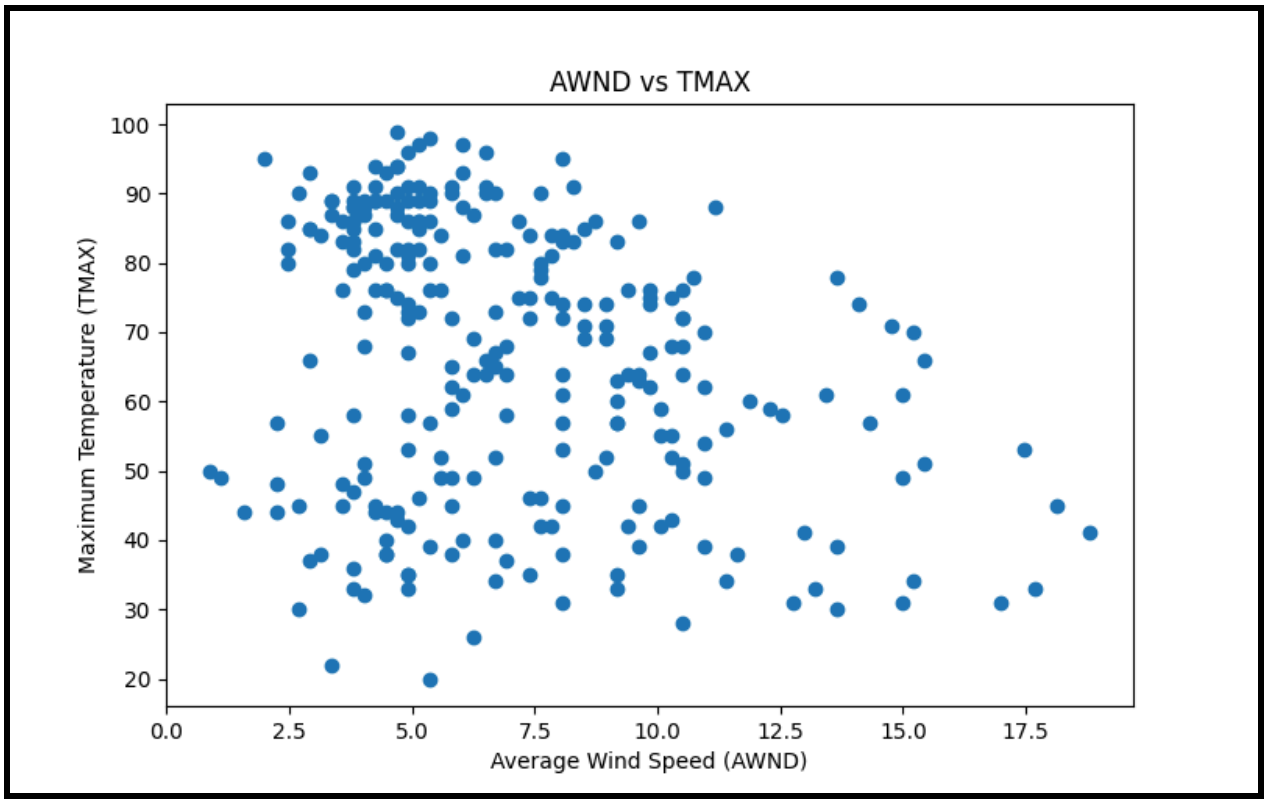
Q8:

1. The correlation coefficient between SNWD and TMAX is -0.28. What this means is that the snow depth and the maximum temperature have a relatively weak inverse relationship. It's obvious that if there is snow, the maximum temperature is already going

to be colder, but it can cause a feedback loop through an albedo effect. The albedo effect occurs when a highly reflective surface, like white snow, bounces incoming short-wave solar radiation back into space. This reduces the amount of heat absorbed by the Earth, ultimately lowering the surrounding air temperature.

2. The correlation coefficient between PRCP and TMIN is 0.17. This means that precipitation and the minimum temperature have a relatively weak direct relationship. Stormy weather can cause a nighttime warming effect through a cloud blanket mechanism. The cloud blanket effect occurs when thick cloud cover, which is required for precipitation, traps outgoing long-wave thermal radiation emitted by the Earth's surface. This prevents heat from escaping completely into space at night, which raises the minimum temperature.
3. The correlation coefficient between TMAX and AWND is -0.29. The maximum temperature and the average wind speed have a relatively weak inverse relationship. This could be the result of a seasonal change during the winter. Cold air naturally creates higher air pressure, while hot air creates lower pressure. Since the overall temperature differences are much larger and sharper during the colder months, it creates a massive imbalance in atmospheric pressure. The atmosphere constantly tries to return to equilibrium by rushing air from high-pressure zones to low-pressure zones, which ultimately triggers much stronger winds on days when temperatures remain low.

F5 - Scatter Plot of AWND vs. TMAX



Q9: From observing the scatter plot of AWND vs. TMAX, there seems to be a weak indirect relationship between them. On days with higher temperatures, there seems to be lower wind speeds. However, since the data is relatively spread out, the relationship is not that strong.

4. Feature Engineering

Q10: Table of Summary Statistics of Engineered Features:

	temp_range	rain_day	wind_gust_ratio
mean	19.26	0.34	4.28
std	7.24	0.47	1.75
min	4	0	1.93
max	40	1	14.18

Q11:

- **temp_range -** It can help identify days with large temperature swings, which may be associated with seasonal patterns, humidity levels, or weather fronts. It may improve prediction models because days with larger temperature ranges often reflect different atmospheric conditions than more stable days.
- **rain_day -** This binary feature indicates whether precipitation occurred. Converting precipitation into a simple yes/no variable can help models detect weather patterns more easily than using raw precipitation amounts alone. It is useful for inference because rainy days are often associated with lower temperatures, higher humidity, stronger winds, and cloudier conditions.
- **wind _gust_ratio -** This feature compares peak wind gust speed to average wind speed. Higher values indicate unusually strong gusts relative to normal wind conditions, which may signal storms or unstable weather conditions. It may improve prediction by capturing wind variability rather than just average wind speed.

Q12: A hot-day indicator can help models identify extreme heat events more clearly than raw temperature values alone. Very hot days may be associated with lower precipitation, different wind patterns, or seasonal effects. Binary indicators are also useful because they simplify interpretation and can highlight threshold-based behavior in the data.

5. Statistical Inference

Q13: TAVG - January vs. July

- **H_0:** $\mu_{July} - \mu_{January} = 0$
- **H_1:** $\mu_{July} - \mu_{January} \neq 0$
- **January mean** ≈ 30.65 Degrees Fahrenheit
- **July mean** ≈ 80.52 Degrees Fahrenheit
- **Difference (July − January)** ≈ 49.87 Degrees Fahrenheit
- **95% CI:** (46.86, 52.88)
- **p-value** = 1.9×10^{-28}
- **Interpretation:** July is about **50°F warmer** than January on average. The statistical test gives an extremely small p-value, which means there is overwhelming evidence that July is truly warmer than January in the underlying population. We reject the null hypothesis.

Q14: Wind Gust Ratio - Rain vs. No Rain

- **H_0:** $\mu_{rain} - \mu_{norain} = 0$
- **H_1:** $\mu_{rain} - \mu_{norain} \neq 0$
- **Rain mean** ≈ 4.65
- **No Rain mean** ≈ 4.10
- **Difference (July − January)** ≈ 0.55
- **95% CI:** (0.03, 1.08)
- **p-value** ≈ 0.0389

- **Interpretation:** Wind gust ratios are about 0.55 higher on rain days compared to non-rain days on average. The statistical test produced a p-value of 0.0389, which is less than 0.05, indicating that this difference is statistically significant. Therefore, we reject the null hypothesis and conclude that wind gust ratios are higher on rainy days than on non-rainy days in the underlying population.

Q15:

- **r** = -0.1898
- **95% CI:** (-0.3021, -0.0723)
- **Interpretation:** This indicates a weak indirect relationship between temperature range and average wind speed. As temperature range increases, average wind speed tends to decrease slightly.

6. Regression Modeling

Q16: Predicted TMAX = β0 + β1(TMIN) + β2(AWND) + β3(rain_day) + β4(wind_gust_ratio)

Q17:

β0: 17.406

β1: 0.996

β2: -0.173

β3: -5.723

β4: 1.052

RMSE: 6.547

MAE: 5.202

R^2: 0.884

Q18:

TMIN appears to be the most important predictor because it has the largest and most direct relationship with TMAX. The coefficient is close to 1, meaning that minimum and maximum daily temperatures are strongly related.

Since it is linear regression, normalization isn't really necessary because the model can still predict accurately even with variables on different scales.

AWND and wind_gust_ratio are overlapping features because the calculation of wind_gust_ratio involves AWND, so these features are not completely independent of each other. This could introduce multicolinearity to the model.

A good set of predictors should include variables that provide unique information about the target variable. Highly correlated or derived predictors can make model interpretation more difficult.

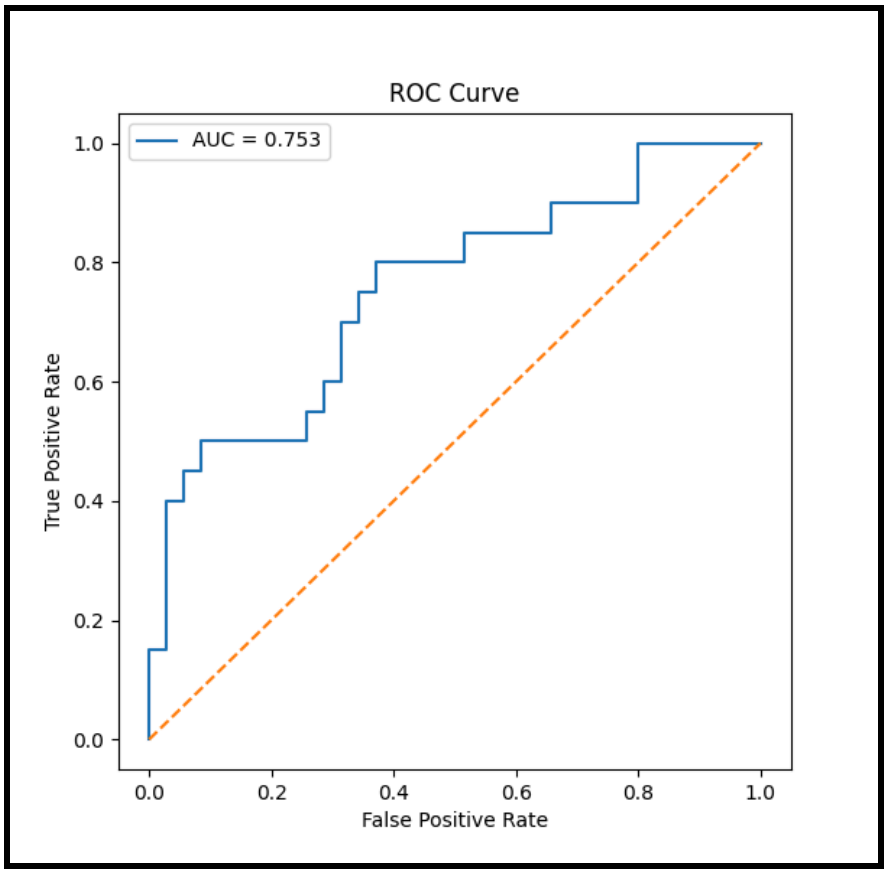
7. Classification Modeling

Q19: Logistic regression was used for the model and the given features: TAVG, temp_range, AWND, and wind_gust_ratio.

Q20:

	Actually Positive	Actually Negative
Predicted Positive	33	2
Predicted Negative	11	9

- **Accuracy** = 76.36 %



- **AUC:** 0.753

Q21:

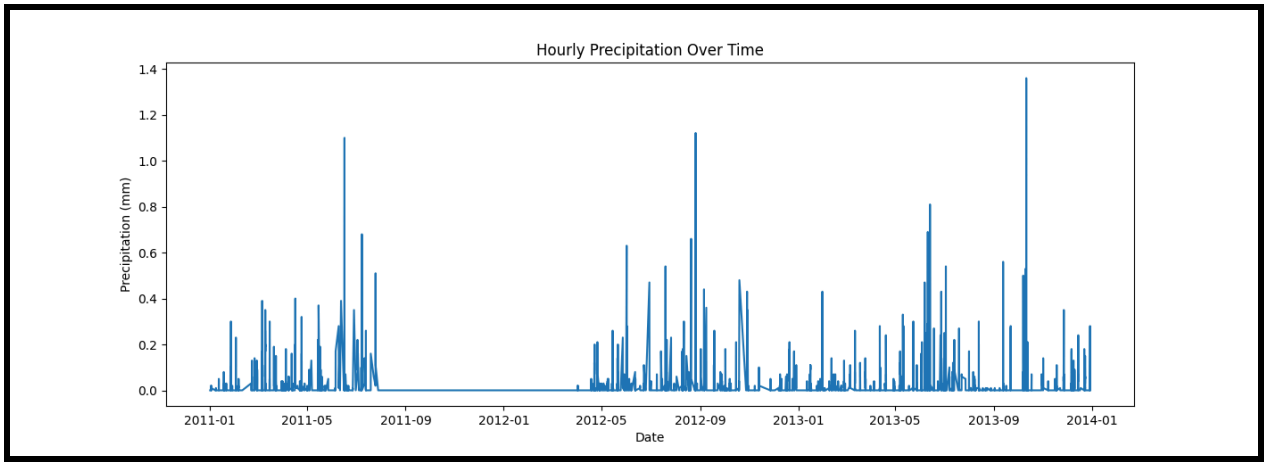
- **TAVG :** 0.021
- **temp_range :** -0.162
- **AWND :** -0.007
- **wind_gust_ratio :** 0.265
- The most predictive variables in the model appear to be wind_gust_ratio and temp_range. Higher wind gust ratios are associated with a greater probability of rain, while larger temperature ranges are associated with a lower probability of rain, likely because rainy days tend to have more cloud cover and smaller temperature swings. TAVG has only a small positive effect, and AWND contributes very little to the model. The classifier makes more false negatives (11) than false positives (2), meaning it is more likely to miss actual rain days than incorrectly predict rain. This suggests the model is somewhat conservative when predicting rainfall.

8. Time Series Analysis — Hourly Precipitation

Q22: After sorting and checking the time series data, I noticed that the time stamps were not spaced out regularly. There were only 2,738 observations when there should have been 26,247 observations, making 23,509 observations missing. Since hourly rainfall can change so quickly, trying to guess and fill in the blanks using technical methods like forward filling or interpolation is risky because it can create a very unrealistic picture of what actually happened. A better approach is to leave the gaps as they are to avoid creating misleading patterns.

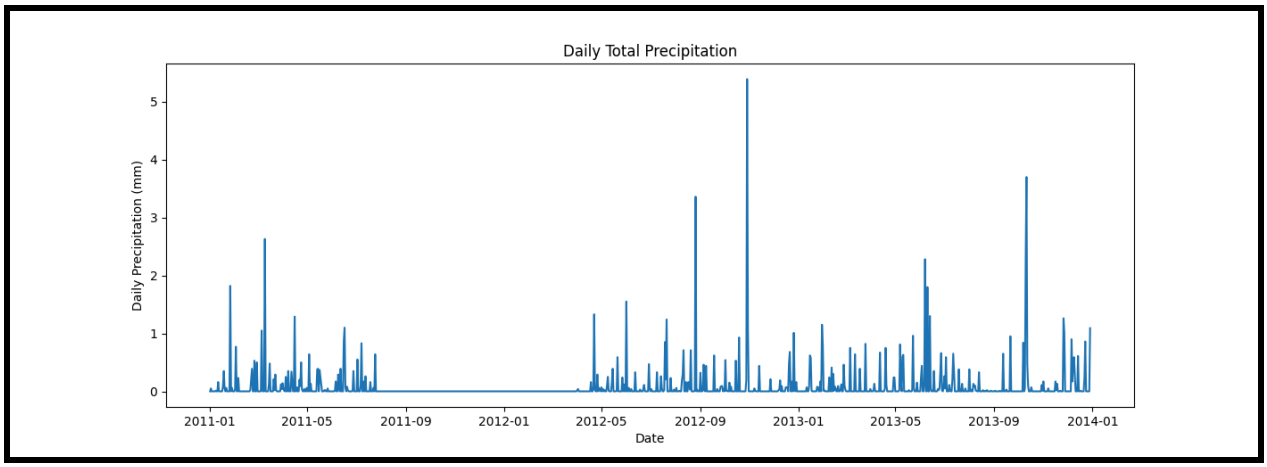
Also, I noticed that between 8/1/2011 1:00:00 and 4/1/2012 0:00:00, the precipitation is 999.99 mm which seems too abnormal to be real considering the other values and the measurement is ‘]’ which is not a valid value. I would remove these observations as they must have been the result of some error.

F6 - Hourly Precipitation Over Time



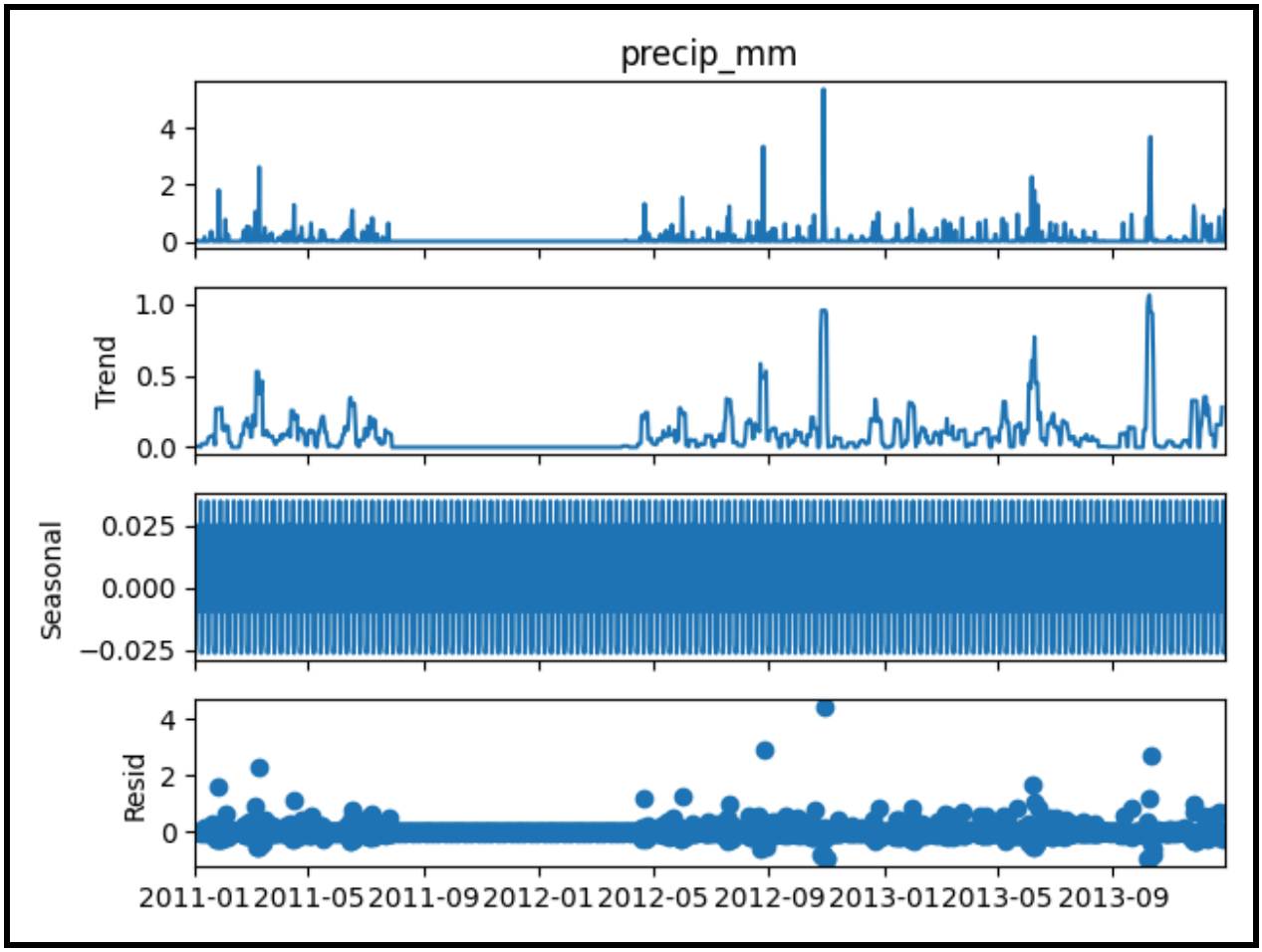
Q23: Looking at the hourly time series graph of precipitation, there seems to be a pattern of large spikes, with the largest ones being around the fall with another one being in the summer. There is also a long list of zeros between July 2011 and May 2012.

F7 - Daily Total Precipitation



Q24: When comparing the daily total precipitation with the hourly precipitation graphs, I see that the overall pattern stays the same. The pattern being spikes and then flat zeroes and then continued spikes. The only difference is the magnitude of the spikes for each part of the graph have changed with the largest one being in October 2012.

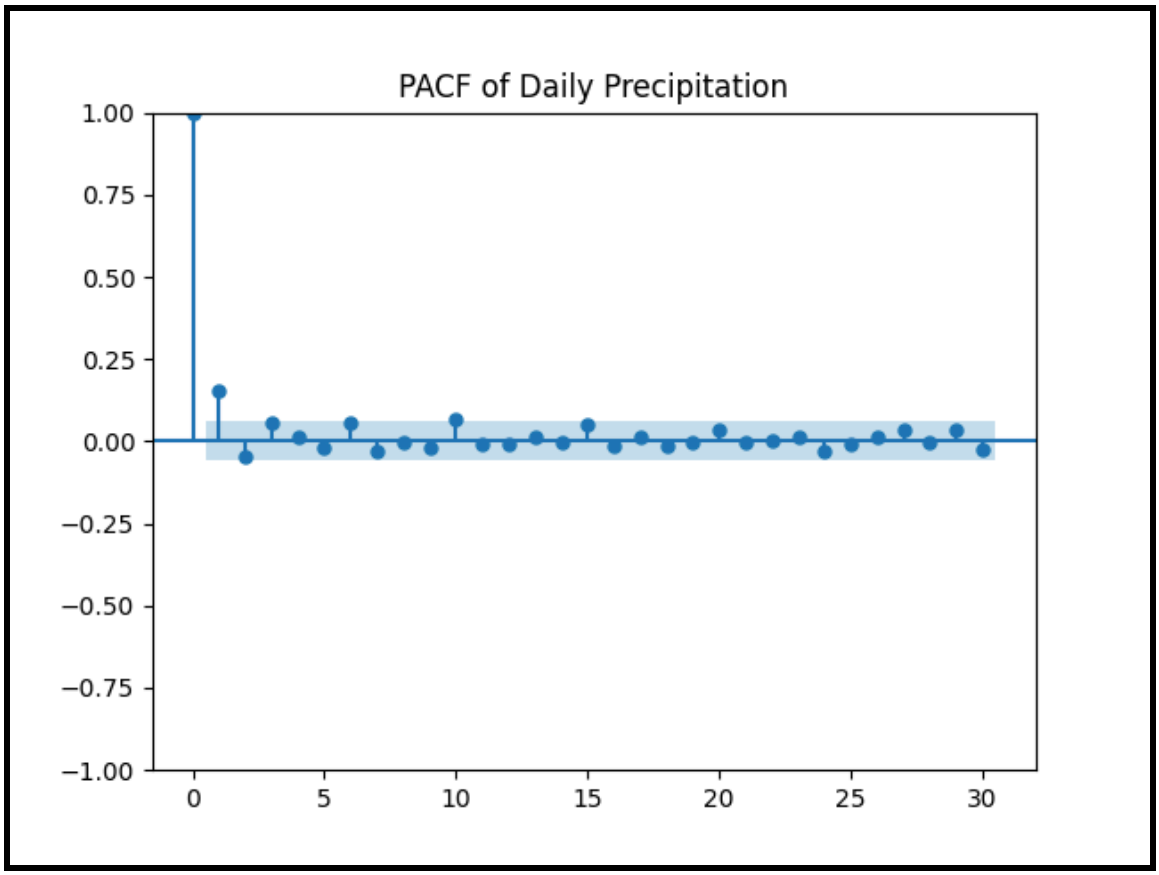
Time Decomposition using Daily Series



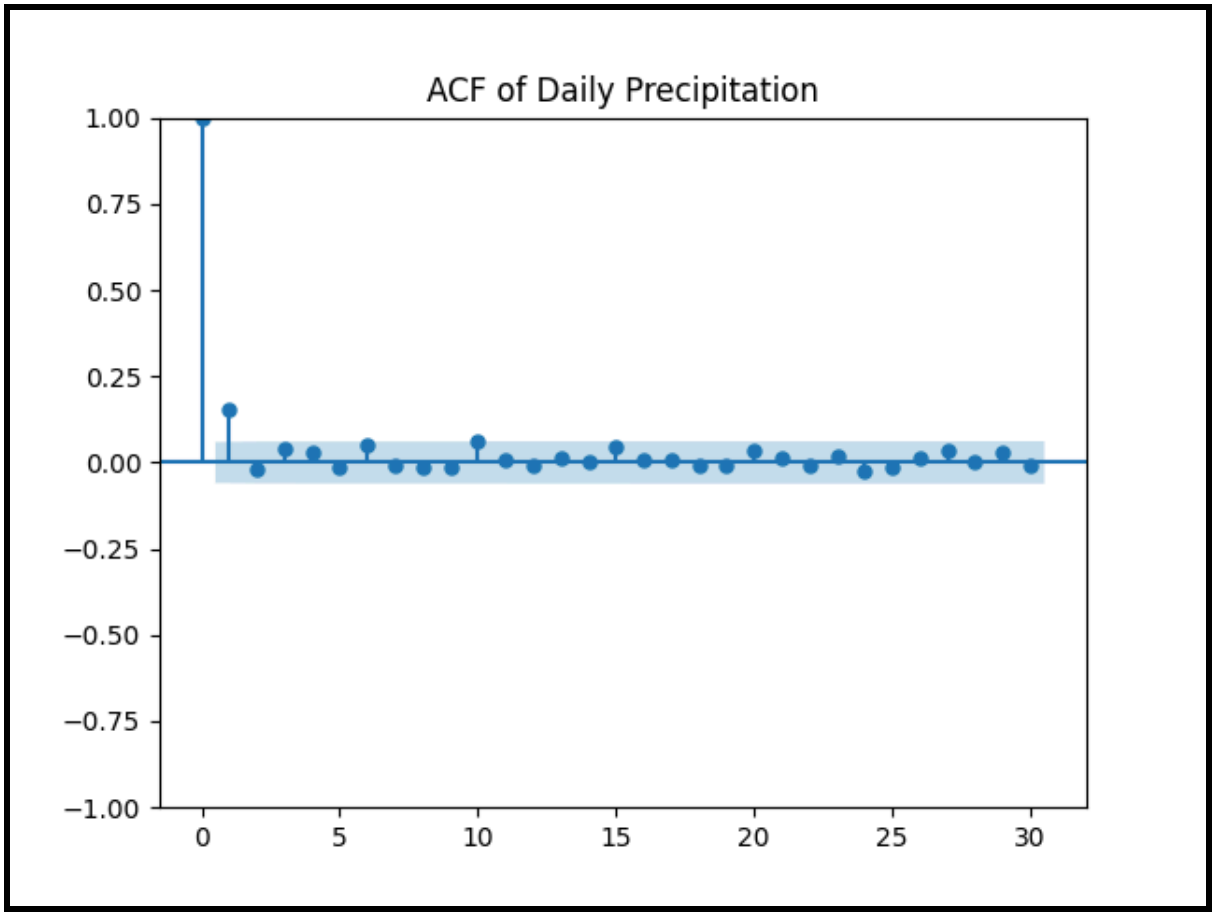
Q25: The decomposition of the daily precipitation series reveals that rainfall is highly sporadic, characterized by a prolonged dry spell spanning late 2011 through early 2012 that is interrupted by sudden, high-magnitude storm spikes. The trend component effectively smooths these daily fluctuations to highlight broader climatic shifts, revealing a baseline of zero during the drought and heightened baseline moisture leading into late 2012 and late 2013. Meanwhile, the seasonal component plays a virtually negligible role, varying by less than absolute value of 0.025 mm and confirming that rainfall in this region is driven by chaotic weather systems rather than a predictable calendar rhythm. Consequently, the bulk of the data's variance is absorbed by the

residual component, where the massive upward outliers vividly isolate extreme, single-day weather anomalies that the smooth trend cannot account for.

PCAF of Daily Precipitation



PCAF of Daily Precipitation



Q26: The only meaningful lag in both ACF and PACF plots is lag 1, which suggests short-term persistence in precipitation where rainfall on one day is somewhat related to rainfall on the following day. Most of the remaining lags fall within the confidence interval, indicating weak long-term autocorrelation and little evidence of strong seasonal behavior in the series.

Q27: AR(1) model was selected because the PACF plot showed the strongest partial autocorrelation at lag 1, while higher lags were much smaller. Since the autocorrelation weakens quickly after lag 1, being within the confidence interval, the series does not appear to have strong long-term memory.

AR coefficients:

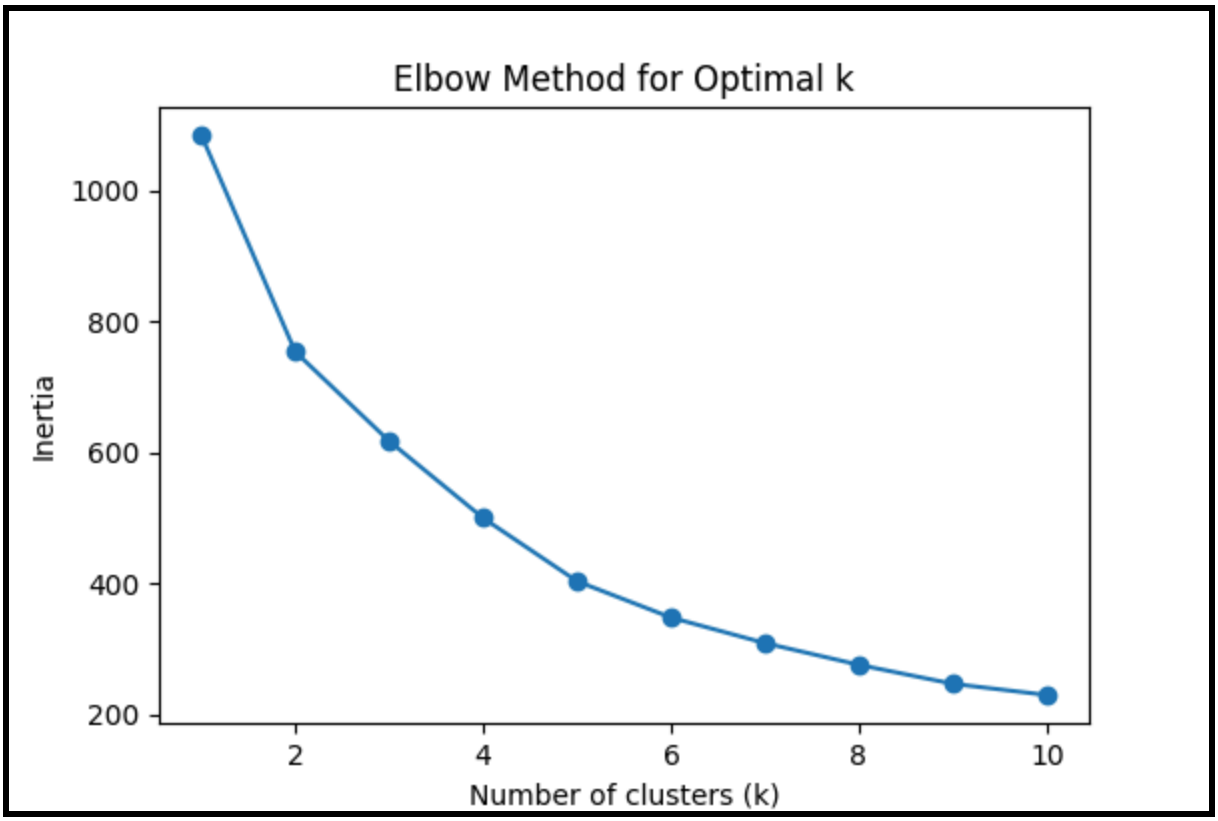
const = 0.0530

AR(1) = 0.1531

Interpretation: The intercept is the baseline precipitation level, and the AR(1) term shows a weak positive persistence, meaning rainfall today is only slightly influenced by yesterday's rainfall.

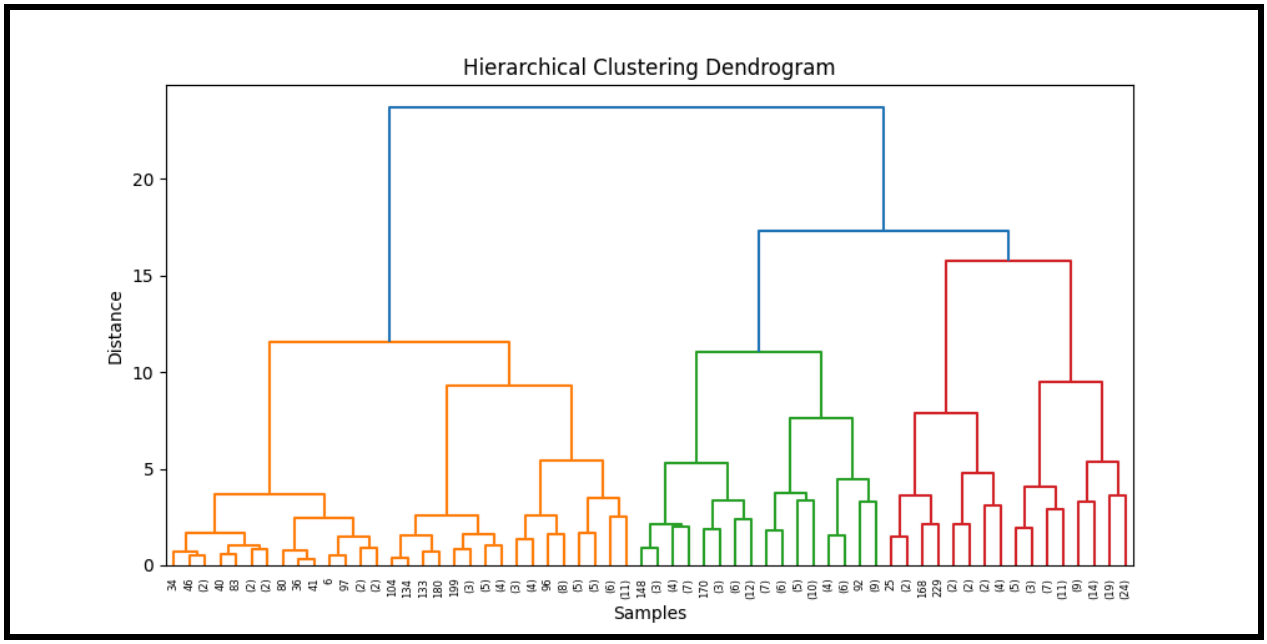
9. Clustering Analysis

Q28: Features used were TAVG, AWND, temp_range, and wind_gust_ratio.

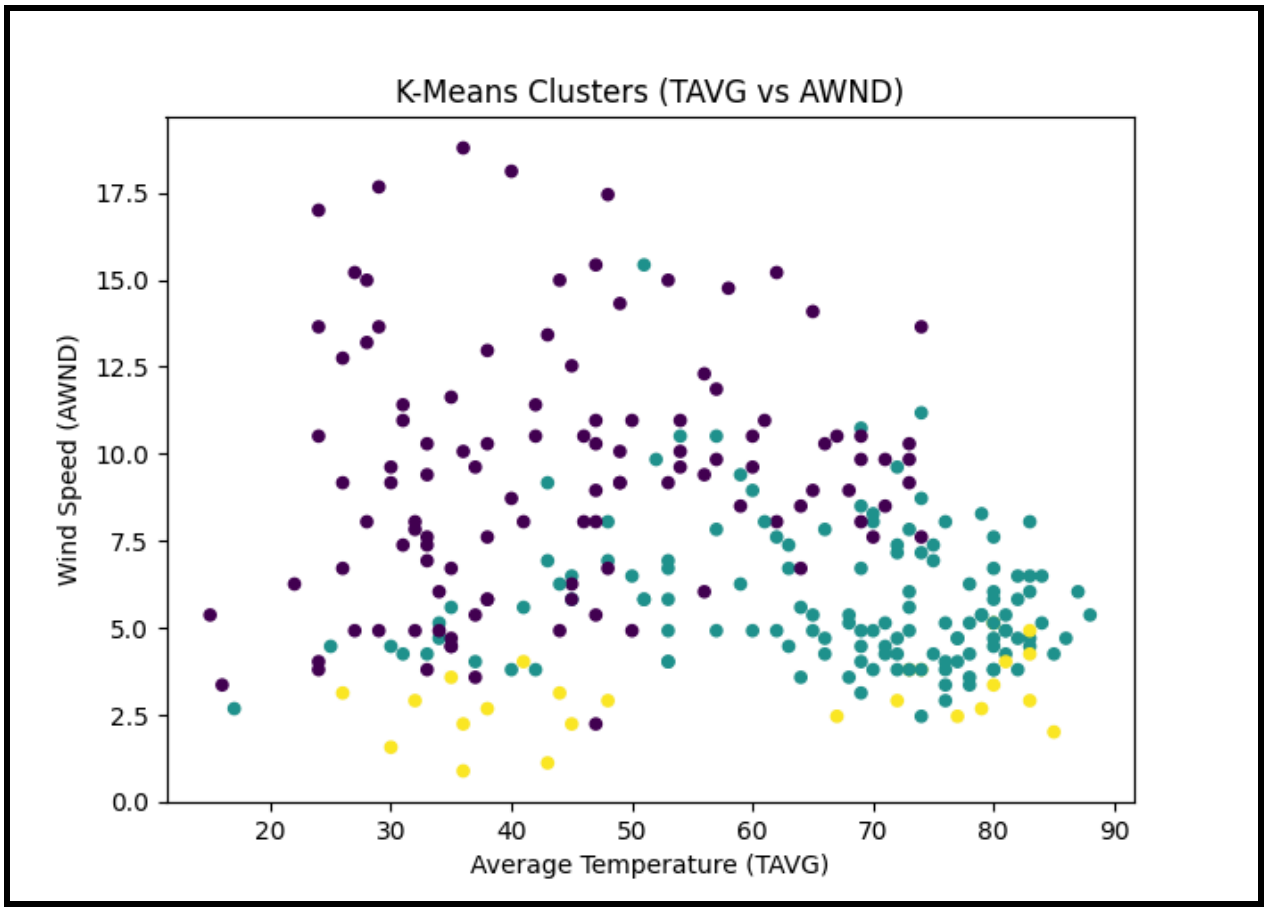


Q29: I selected k = 3 using the elbow method by examining the inertia plot. The reduction in inertia slows noticeably at k = 3, indicating diminishing returns beyond that point, so I chose k = 3 as the optimal number of clusters.

F8.1 - Dendrogram



F8.2 - Scatter Plot



Q30: The turquoise cluster represents hot conditions with relatively low wind, suggesting calm, warm weather days. The yellow cluster corresponds to cooler temperatures with low wind speeds, indicating stable, cooler weather conditions. The purple cluster shows moderate temperatures combined with higher wind speeds, which is consistent with windier or more active weather systems. Overall, the clusters appear to reflect different weather regimes driven primarily by temperature, with wind acting as a secondary factor distinguishing between otherwise similar conditions. These clusters broadly align with seasonal patterns, separating

warmer summer-like conditions, cooler winter-like conditions, and transitional or variable weather periods.

10. AI Usage Statement

ChatGPT was used to support implementation of Python libraries, machine learning models, and code for making plots while Gemini was used to assist with grammar and writing refinement in my report.